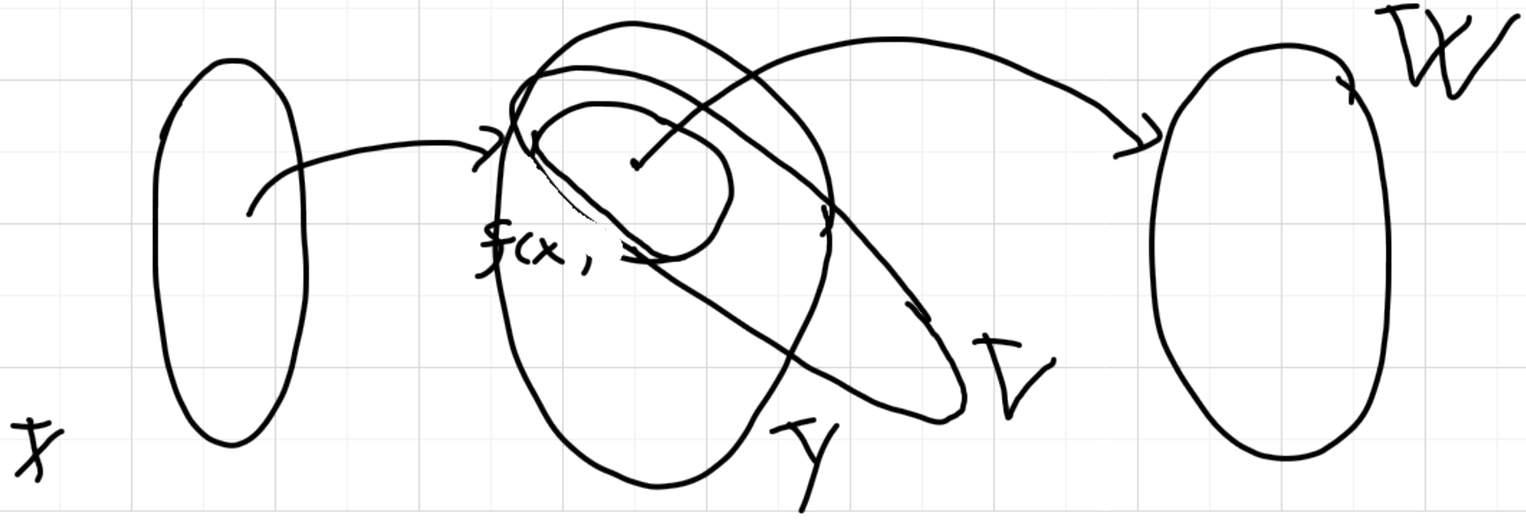
Def.

Siano X, Y, V, W insiemi e

$$f : X \to Y$$
$$g : V \to W$$

la funzione $f \circ g : X \to W$



$$\text{Dom}\{f \circ g\} = \{x \in X : g(x) \in \text{Dom } f\}$$

- $f \circ g \neq g \circ f$ ⊗ NO CONMUTATIVA

- $f \circ (g \circ h) = (f \circ g) \circ h$ ✓ SI ASSOCIATIVA

$$f(x) = \sqrt[3]{x} \qquad g(x) = x^2 - 1$$

$$f \circ g = f(g(x)) = f(x^2 - 1) = \sqrt[3]{x^2 - 1}$$

$$g \circ f = g(f(x)) = g\left(\sqrt[3]{x}\right) = \left(\sqrt[3]{x}\right)^2 - 1 =$$

$$= x^{\frac{2}{3}} - 1$$

Limiti di successione

Def -

La successione di numeri reali è una funzione $a: \mathbb{N} \to \mathbb{R}$

$n \longmapsto a_n$

n ← indice; a_n → termine della successione

a: dominio

"puntiforme"

COSA VOGLIAMO SAPERE?

$\{a_n\}$

- È LIMITATA
  - SUPERIORMENTE $a_n \leq M \quad \forall n \in$
  - INFERIORMENTE $a_n \geq M \quad \forall n \in$
  - SUP. AND INFER. $|a_n| \leq M$
- È (STRETTAMENTE) CRESCENTE $a_n \leq (<) a_{n+1}$
- È (STRETT.) DECRESCENTE $a_n \geq (>) a_{n+1}$

COME FACCIAMO ?

$\lim_{n\to\infty} a_n$

- CONVERGE A $\ell \in \mathbb{R}$ ($\ell = 0$ INFINITESIMA)
- DIVERGE A $+\infty$
- DIVERGE A $-\infty$
- $\nexists$? $\Rightarrow$ FORME INDETERMINATE (es. $a_n = (-1)^n$)

$\frac{\infty}{\infty}$ $\quad +\infty - \infty$ $\quad \frac{0}{0}$ $\quad 0 \cdot \infty$

$1^{\pm\infty}$ $\quad (+\infty)^0$ $\quad 0^0$

Gerarchia di infiniti

$$\log_a n \ll n^a \ll a^n \ll n!$$

$$\lim_{n\to\infty} \frac{\log_a n}{n^a} = 0$$

$$\lim_{n\to\infty} \frac{n^a}{a^n} = 0$$

- - - -

Ese 1.

$$\lim_{n\to\infty}\left(2+\frac{5}{n}\right) = \lim_{n\to\infty} 2 + \lim_{n\to\infty}\frac{5}{n}$$

$$= 2 + 5\cdot\lim_{n\to\infty}\frac{1}{n} = 2+5\cdot 0$$

$$= 2$$

Ese 2.

$$\lim_{n\to\infty}\frac{3n^2-2n+1}{n^2+4n-7} = \lim_{n\to\infty}\frac{n^2\left(3-\frac{2}{n}+\frac{1}{n^2}\right)}{n^2\left(1+\frac{4}{n}-\frac{7}{n^2}\right)} =$$

($\frac{2}{n}\to 0$, $\frac{1}{n^2}\to 0$, $\frac{4}{n}\to 0$, $\frac{7}{n^2}\to 0$)

$$= \frac{3}{1} = 3$$

Ese 3

$(a-b)(a+b)=a^2-b^2$ | $\infty-\infty$

$$\lim_{n\to\infty} \sqrt{n^2-n} - n = \lim_{n\to\infty} \left(\sqrt{n^2-n} - n\right) \cdot \frac{\sqrt{n^2-n} + n}{\sqrt{n^2-n} + n} =$$

$$= \lim_{n\to\infty} \frac{n^2 - n - n^2}{\sqrt{n^2\left(1-\frac{1}{n}\right)} + n} = \lim_{n\to\infty} \frac{-n}{n\sqrt{1-\frac{1}{n}} + n} =$$

$$= \lim_{n\to\infty} \frac{n(-1)}{n\left(\sqrt{1-\frac{1}{n}} + 1\right)} \quad \left(\frac{1}{n}\to 0\right) = -\frac{1}{2}$$

Ese 4.

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = \lim_{t\to 0} \frac{\sin t}{t} = 1$$

$$t = \frac{1}{n}$$

$$t \to 0 \quad \text{per } n \to \infty$$

Ese 5.

$$\lim_{n\to\infty} \frac{n^3 + 2^n}{n!} = \lim_{n\to\infty} \frac{n^3}{n!} + \frac{2^n}{n!} = 0$$

$\frac{n^3}{n!} \to 0$

$\frac{2^n}{n!} \to 0$

Ese 6.

$$\lim_{n\to\infty} \frac{4n^2-1}{2n^2+3} + \frac{\log n}{n} = \frac{4}{2} + 0 = 2$$

$\log n$ << n

$$\frac{1}{\infty} = 0$$

Ese 7.

$$(a^3-b^3)=(a-b)(a^2+ab+b^2)$$

$$a-b=\frac{a^3-b^3}{a^2+ab+b^2}$$

$$\lim_{n\to\infty} \underbrace{\sqrt[3]{n^3-n^2}}_{a} - \underbrace{n}_{b} =$$

$$\lim_{n\to\infty} \frac{\cancel{n^3}-n^2-\cancel{n^3}}{\left(\sqrt[3]{n^3-n^2}\right)^2+n\cdot\sqrt[3]{n^3-n^2}+n^2} =$$

$$= \lim_{n\to\infty} \frac{-n^2}{\left(n\cdot\sqrt[3]{1-\frac{1}{n}}\right)^2+n\cdot n\sqrt[3]{1-\frac{1}{n}}+n^2} =$$

$$= \lim_{n\to\infty} \frac{\cancel{n^2}(-1)}{\cancel{n^2}\left[\left(\sqrt[3]{1-\frac{1}{n}}\right)^2+\sqrt[3]{1-\frac{1}{n}}+1\right]} = -\frac{1}{3}$$

($\frac{1}{n}\to 0$ in both cube roots)

$$\sqrt[3]{n^3-n^2}=\sqrt{n^3\left(1-\frac{1}{n}\right)}$$

$$=n\sqrt{1-\frac{1}{n}}$$

Ese 8

$$\lim_{n\to\infty} \underbrace{\sqrt[3]{n+1}}_{a} - \underbrace{\sqrt[3]{n-1}}_{b} =$$

$$= \lim_{n\to\infty} \frac{n+1-(n-1)}{\left(\sqrt[3]{n+1}\right)^2 + \sqrt[3]{(n+1)(n-1)} + \left(\sqrt[3]{n-1}\right)^2} =$$

$$= \lim_{n\to\infty} \frac{2}{\infty} = 0$$